

FBISE Biology Class 9 Model Paper 2026 — Complete Solution

The paper follows the new FBISE format: **50% MCQs and 50% descriptive questions**

Section A — MCQs

Detailed reasoning

1. **C — Living organisms and their interactions with nature.**
Biology studies life, living organisms and their relationship with the environment. A is wrong because biology depends on observation and experimentation. B is too limited, while D restricts biology to history.
2. **A — It encourages observation and study of nature.**
The Quran repeatedly invites humans to observe and reflect upon living things and natural phenomena. The other choices incorrectly limit nature to resources, history or appreciation without investigation.
3. **B — Study of drugs and their effects.**
Pharmacology studies drugs, their actions and effects. Defense against germs is immunology; diseases are studied in pathology; body form and function are studied in anatomy and physiology.
4. **A — Computer science.**
Computer science helps analyse biological data, model protein structures and process vaccine-trial results quickly. Physics and mathematics may also assist research, but computer science is the most directly useful option for fast and accurate data analysis. Geography is not the best choice.
5. **C — Genetics.**
Genetics studies genes, mutations and inheritance. Pharmacology concerns drugs, immunology concerns defense, and physiology concerns body functions.
6. **D — Theory.**
A scientific theory is a well-tested, evidence-supported explanation that can be revised when new evidence appears. A law describes a consistent relationship; an observation is a recorded fact; a hypothesis is a testable tentative explanation.
7. **A — Fungi.**
Fungi are eukaryotic, possess cell walls and absorb dissolved nutrients. Plants mainly prepare food through photosynthesis; animals ingest food; Protista contains organisms with several nutritional modes.
8. **C — Variety of living organisms in an area or on Earth.**
This is biodiversity. A describes ecology, B describes classification, and D describes identification and nomenclature.
9. **C — Genus.**
The hierarchy is:

Kingdom → Phylum/Division → Class → Order → Family → Genus → Species

Therefore, genus comes immediately after family.

10. D — Physical characteristics.

Early classification depended mainly on visible features such as shape, habitat and mode of movement. DNA sequencing and genetic analysis are modern methods.

11. B — Smooth endoplasmic reticulum.

Smooth ER synthesizes lipids and helps detoxify harmful substances. Rough ER carries ribosomes and mainly makes proteins; Golgi modifies and packages materials; chloroplast performs photosynthesis.

12. A — Nerve cells.

Neurons receive and transmit electrical impulses for coordination. Muscle cells produce movement, epithelial cells cover surfaces, and blood cells perform transport and defense.

13. B — Nucleus controlling cellular activities.

Different organelles perform specialised jobs within a cell; this is cellular division of labour. The other examples involve reproduction or activities at tissue and organ levels.

14. The label pointing to the nucleolus.

Ribosomal RNA is produced and ribosomal subunits begin assembling in the nucleolus—the dark, dense body inside the nucleus. Labels pointing to the nuclear envelope, chromatin or nucleoplasm are incorrect.

15. Select the organelle not directly involved in protein synthesis or modification.

Ribosomes synthesize proteins, rough ER transports newly formed proteins and Golgi apparatus modifies and packages them. Therefore, the diagram showing an unrelated organelle—such as a mitochondrion—is the exception.

16. B — Cell membrane absent in plant cells.

This row is incorrect because plant cells possess a cell membrane beneath the cell wall. Centrioles are usually absent in higher plant cells, cell walls support plants, and chloroplasts perform photosynthesis.

17. A — Genetic stability.

Meiosis directly supports sexual reproduction, creates genetic variation and maintains the chromosome number across generations. “Genetic stability” is not its main distinguishing importance because meiosis deliberately generates new combinations.

18. A — Prophase.

During prophase, chromosomes condense and the nuclear membrane begins disappearing. In metaphase chromosomes align centrally; in anaphase they separate; in telophase new nuclei form.

19. B — Ensures DNA replication.

During the S phase of interphase, DNA is duplicated. Cell division occurs in the M phase, while nuclear-membrane breakdown and spindle formation occur during prophase.

20. D — G₀ phase.

Cells that temporarily or permanently stop dividing enter G₀. G₁ is growth, S is DNA synthesis and G₂ is preparation for division.

21. B — Repair of damaged tissues by forming new cells.

Stem cells are unspecialised cells that can divide and differentiate. They do not primarily transport oxygen, digest food or deliberately cause mutations.

22. **Select the diagram showing homologous chromosomes moving to opposite poles while sister chromatids remain joined.**
That is anaphase I. A diagram showing sister chromatids separating at their centromeres represents mitotic anaphase or anaphase II.
23. **D — Hypothalamus.**
The hypothalamus acts as the body's temperature-control centre. Lymph nodes support immunity, the pituitary releases hormones, and kidneys regulate water and salts.
24. **B — Maintain a stable internal environment.**
Homeostasis keeps temperature, water, salts, glucose and pH within suitable limits. It does not mean that the body changes rapidly with the environment.
25. **D — Brain: organ.**
The brain consists of several tissues working together, so it is an organ. A root is an organ, the heart is an organ and "muscle" generally refers to tissue rather than an organ system.
26. **B — Emergent properties at the cellular level.**
A complete cell can perform functions that one organelle cannot. New abilities appear when organelles interact; these are emergent properties.
27. **B — Organelle → Cell → Tissue → Organ → System → Organism.**
Each level combines to form the next higher organisational level.
28. **B — Mechanical digestion.**
Contraction of stomach muscles churns and mixes food. Enzymes carry out chemical digestion; nutrient absorption mainly occurs in the small intestine.
29. **A — Gametes.**
Sperm and egg cells carry hereditary information to the next generation. A zygote forms after their fusion; stem and brain cells normally do not transmit genes to offspring.
30. **A — Act as enzymes.**
Most enzymes are proteins. Lipids store energy, provide insulation and form an important part of cell membranes.
31. **C — Molecules present in cells and their reactions.**
Molecular biology studies DNA, RNA, proteins and their interactions. Organ systems belong to anatomy and physiology; microorganisms are studied in microbiology.
32. **C — Instant source of energy.**
Carbohydrates are broken down into glucose for respiration. They do not mainly transport gases, provide immunity or act as enzymes.
33. **A — Milk, eggs and fish.**
These are rich sources of protein. Fruits and vegetables mainly supply vitamins and minerals; butter and oil mainly supply lipids; cereals and potatoes mainly supply carbohydrates.
34. **B — Glycogen.**
Excess glucose is stored as glycogen, especially in the liver and muscles. Cellulose is a plant carbohydrate; vitamins are not glucose stores.
35. **B — Guanine ≡ Cytosine.**
Guanine and cytosine form three hydrogen bonds. Adenine pairs with thymine through two hydrogen bonds.

36. **B — Active transport.**
Active transport moves substances against a concentration gradient and requires ATP. Osmosis, diffusion and facilitated diffusion are passive processes.
37. **A — Large surface area of leaves.**
Hydrophytes live where water is abundant and commonly possess broad leaves. Leaf shedding, closed stomata and spiny leaves are mainly water-conserving adaptations of xerophytes.
38. **C — Cuttings and grafting.**
These are asexual methods. Megaspore formation, microspore formation and double fertilisation are associated with sexual reproduction.
39. **D — Iron and manganese.**
Plants require these elements in very small quantities. Carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur are macronutrients.
40. **C — It involves mitosis in the parent body.**
Mitosis produces genetically identical cells, so a bud normally becomes a clone of its parent. Meiosis and cross-pollination cause or contribute to variation.
41. **B — Optimum temperature.**
At 37°C the enzyme has its maximum activity. It is not the freezing point or a point of substrate loss.
42. **C — Enzymes were inactive.**
Low temperature greatly reduces molecular movement and enzyme-substrate collisions. The enzymes are not normally denatured at 0°C; they become active again after warming.
43. **B — Denaturation of enzyme.**
Above the optimum temperature, bonds maintaining the active site are disrupted. The active site changes shape, so the reaction rate falls sharply.
44. **A — No significant change in reaction rate.**
At 50°C the human enzyme is largely denatured. Adding more enzyme will not permanently restore activity because the newly added enzyme is exposed to the same damaging temperature.
45. **B — High rate of transpiration.**
In hot, dry air, water loss may exceed water absorption. Cells lose turgor and leaves wilt.
46. **C — Stomata close, reducing CO₂ intake.**
Closing stomata conserves water but restricts carbon dioxide entry, lowering photosynthesis.
47. **B — Pressure-flow mechanism depends on water balance.**
Phloem translocation requires water to produce hydrostatic-pressure differences between sources and sinks.
48. **A — Xylem and phloem.**
Xylem transports water and mineral salts; phloem transports sugars and other organic solutes.
49. **B — Aerobic respiration.**
High oxygen, high ATP and no lactic-acid accumulation indicate aerobic respiration.
50. **B — Active exercising muscle.**
During vigorous exercise oxygen supply may become insufficient, so muscle cells perform lactic-acid fermentation.

51. **A — Cell A is performing aerobic respiration.**

Cell B, with low oxygen and lactic acid, performs anaerobic respiration.

52. **B — Oxygen is not used for complete oxidation.**

In Cell B, glucose is only partially broken down, so much of its energy remains in lactic acid and only a small amount of ATP is produced.

53. **B — Vegetative propagation through layering.**

The attached stem develops roots after being covered with soil. Strictly, deliberate layering by a gardener is an artificial vegetative-propagation technique, but B is the intended and best available option.

54. **D — Artificial propagation through cutting.**

A detached stem develops roots and shoots to produce a new plant.

55. **B — Natural vegetative propagation occurs naturally; artificial propagation requires human involvement.**

Both are asexual and usually produce genetically similar offspring. Artificial propagation is not restricted to laboratories.

56. **C — Both vegetative and artificial propagation can produce identical plants.**

Both depend on mitotic division and can produce clones. Identical plants do not prove that a mutation or sexual reproduction occurred.

57. **B — Vestigial structure.**

The appendix is reduced compared with the corresponding functional structure in ancestors.

58. **B — Evolution has occurred over generations.**

Vestigial structures provide evidence of descent with modification.

59. **D — They lost their major function over generations.**

Environmental and lifestyle changes made their original functions less important. They were not artificially created or newly evolved.

60. **A — Hind-limb bones of snakes.**

These reduced bones are remnants of limbs possessed by ancestors. The heart, wings and gills are functional structures.

Section B — Short-Response Questions

Question 1

Binomial nomenclature

Binomial nomenclature is the internationally accepted system in which every species is assigned a two-word scientific name:

1. The first word is the **genus** and begins with a capital letter.
2. The second word is the **specific epithet** and begins with a small letter.
3. Both words are italicised when printed or separately underlined when handwritten.

Examples:

- Animal: *Homo sapiens*
- Plant: *Mangifera indica*

OR: Identification of the plant structure

X is a flower, the reproductive structure of an angiosperm.

The labels should be matched with the arrows in the supplied diagram:

- **Stigma:** receives pollen grains.
- **Anther:** produces pollen grains.
- **Ovary:** contains ovules.
- **Ovule:** contains the female gametophyte and develops into a seed after fertilisation.
- **Petal:** attracts pollinators.
- **Sepal:** protects the flower bud.

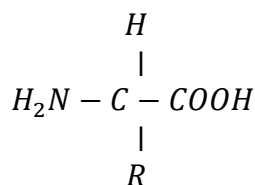
In the examination, write the four names that correspond exactly to arrows A–D.

Question 2

Primary structure of proteins

The primary structure of a protein is the **specific number, type and linear sequence of amino acids joined by peptide bonds in a polypeptide chain.**

General structure of an amino acid



Where:

- NH_2 = amino group
- $COOH$ = carboxyl group
- H = hydrogen atom
- R = variable side chain

Role of proteins in defense

Antibodies are defensive proteins produced by B-lymphocytes. Each antibody recognises a particular antigen and helps neutralise or destroy the pathogen carrying it.

OR: Characteristics of a good hypothesis

A good hypothesis must be:

1. **Testable:** It can be examined through an experiment.
2. **Falsifiable:** Evidence could show it to be incorrect.
3. **Based on observations and existing knowledge:** It must have a logical scientific basis.
4. **Clear and specific:** Variables and their predicted relationship should be understandable.
5. **Predictive:** It should lead to deductions that can be tested.

Any three well-explained characteristics earn full marks.

Question 3

Macronutrients and micronutrients

Macronutrients

Required in comparatively large quantities.
Examples: nitrogen, phosphorus, potassium, calcium, magnesium and sulphur.

Micronutrients

Required in trace or very small quantities.
Examples: iron, manganese, zinc, copper, boron and molybdenum.

Role of magnesium: It is a central component of chlorophyll and is therefore essential for photosynthesis.

Role of nitrogen: It is required for amino acids, proteins, nucleic acids and chlorophyll. It promotes healthy vegetative growth.

OR: A.F.A. King's observations about malaria

A.F.A. King observed that:

1. Malaria was common in areas where mosquitoes were abundant, particularly near marshes.
2. People who slept outdoors without protection were more frequently affected.
3. Individuals using mosquito nets were comparatively protected from malaria.
4. Malaria cases increased during seasons in which mosquito populations increased.
5. Measures that reduced mosquitoes also reduced malaria cases.

These observations supported the idea that mosquitoes transmit malaria.

Question 4

Differences between DNA and RNA

DNA	RNA
Usually double-stranded and forms a double helix.	Usually single-stranded.
Contains deoxyribose sugar.	Contains ribose sugar.
Bases are adenine, thymine, guanine and cytosine.	Bases are adenine, uracil, guanine and cytosine.
Mainly stores hereditary information.	Mainly helps express genetic information during protein synthesis.
More chemically stable.	Comparatively less stable.

Any three correct structural differences earn full marks.

OR: Relationship among variation, adaptation and evolution

Variation means differences among individuals of the same species. Variations arise through mutation, crossing over, independent assortment and random fertilisation.

Some variations improve survival and reproduction in a particular environment. Natural selection favours individuals carrying these useful variations. When an inherited beneficial variation becomes common, it is called an **adaptation**.

Over many generations, useful adaptations accumulate and populations change. This gradual inherited change is **evolution**. If populations become sufficiently different and reproductively isolated, new species may form.

Question 5

Angiospermic seed

A labelled dicot seed diagram should include:

- Seed coat/testa
- Cotyledons
- Embryonic axis
- Plumule
- Epicotyl
- Hypocotyl
- Radicle
- Hilum
- Micropyle

Functions:

- **Seed coat:** protection
- **Cotyledons:** stored food
- **Plumule:** develops into the shoot
- **Radicle:** develops into the root
- **Micropyle:** permits entry of water during germination
- **Hilum:** scar showing where the seed was attached to the fruit

OR: Prophase I of meiosis

Important events include:

1. Homologous chromosomes pair through **synapsis**, forming bivalents or tetrads.
2. Non-sister chromatids exchange corresponding segments through **crossing over**.
3. Chiasmata become visible.
4. Chromosomes condense and become shorter and thicker.
5. The nuclear envelope and nucleolus disappear.
6. Spindle fibres begin forming.

For the diagram, show paired homologous chromosomes, each consisting of two sister chromatids, with a chiasma between non-sister chromatids.

Question 6

Anaphase I of meiosis versus anaphase of mitosis

Anaphase I of meiosis	Anaphase of mitosis
Homologous chromosomes separate.	Sister chromatids separate.
Centromeres do not divide.	Centromeres divide.
Each chromosome still consists of two sister chromatids.	Each separated chromatid becomes a daughter chromosome.
Chromosome number is reduced from diploid to haploid.	Chromosome number remains unchanged.
Daughter chromosome sets may be genetically different.	Daughter chromosome sets are normally identical.

In the diagrams:

- **Anaphase I:** draw X-shaped chromosomes moving to opposite poles.
- **Mitotic anaphase:** draw separated V-shaped daughter chromosomes moving to opposite poles.

OR: Pepsin and trypsin pH graph

- **Peak A at approximately pH 2 represents pepsin.** Pepsin works best in the highly acidic environment produced by hydrochloric acid in the stomach.
- **Peak B at approximately pH 8 represents trypsin.** Trypsin works best in the alkaline medium of the small intestine.

Sites of action:

- Pepsin: stomach
- Trypsin: duodenum/small intestine

A major change away from the optimum pH alters ionic interactions in the enzyme, changes its active site and reduces its activity.

Section C — Extended-Response Questions

Question 7

(a) Roles of tissues and structures in a leaf

Identify each arrow from its appearance:

- **Palisade mesophyll:** tightly packed columnar cells beneath the upper epidermis. It contains many chloroplasts and is the main site of photosynthesis.
- **Spongy mesophyll:** loosely arranged cells with air spaces. It permits rapid diffusion of carbon dioxide, oxygen and water vapour and also performs some photosynthesis.
- **Stoma and guard cells:** the stomatal pore permits gaseous exchange and transpiration. Guard cells regulate opening and closing.
- **Xylem:** transports water and mineral salts to the leaf.
- **Phloem:** transports sugars from the leaf to growing or storage regions.
- **Epidermal tissue:** protects the leaf; the waxy cuticle reduces water loss.

Match A, B and C to these descriptions according to their arrows.

(b) Respiratory and urinary systems in homeostasis

Respiratory system

1. Removes carbon dioxide produced during cellular respiration.
2. Carbon dioxide reacts with water to form carbonic acid; therefore, its removal prevents a dangerous fall in blood pH.
3. Gas exchange in alveoli maintains suitable oxygen and carbon-dioxide concentrations.
4. Exhalation also removes some water vapour and heat.

Urinary system

1. Kidneys remove nitrogenous waste, especially urea.
2. Nephrons regulate the amount of water reabsorbed into blood.
3. Kidneys regulate salts and ions such as sodium and potassium.
4. Selective reabsorption prevents loss of glucose and other useful substances.
5. Hydrogen-ion secretion and bicarbonate regulation help maintain blood pH.

Thus, both systems keep the internal environment stable.

Question 7 — OR

(a) Chemical composition and roles of labelled cellular components

Use the following mapping according to the arrows in the supplied diagram.

Cell membrane

Composition: Phospholipid bilayer containing proteins, cholesterol and surface carbohydrates.

Role: Forms a selectively permeable boundary, regulates movement of substances, supports cell recognition and receives chemical signals.

Nucleus

Composition: DNA associated with proteins, RNA, nucleoplasm and a double phospholipid nuclear envelope containing pores.

Role: Stores hereditary information and controls cellular activities by regulating gene expression and protein synthesis.

Mitochondrion

Composition: Double phospholipid membrane, protein-rich inner membrane, enzymes, mitochondrial DNA and ribosomes.

Role: Carries out aerobic respiration and produces ATP.

Cell wall, if labelled

Composition: Primarily cellulose with hemicellulose and pectin.

Role: Provides strength, shape and protection and prevents bursting in a hypotonic environment.

(b) Production and functions of lysosomes

Lysosomal enzymes are proteins synthesised by ribosomes on rough ER. They pass through the ER in transport vesicles and enter the Golgi apparatus. The Golgi modifies and packages them into membrane-bound vesicles called lysosomes.

Autophagy: A lysosome fuses with a vesicle containing a worn-out organelle. Hydrolytic enzymes break down the organelle, and useful products are recycled.

Intracellular digestion: Lysosomes fuse with food vacuoles or vesicles containing microorganisms. Their enzymes digest food particles and destroy many pathogens.

Lysosomal membranes isolate these powerful enzymes and protect the cytoplasm.

Question 8

Aerobic respiration

The three principal stages are glycolysis, the Krebs cycle and the electron transport chain.

X — Glycolysis

- **Site:** Cytoplasm
- **Reactant:** One six-carbon glucose molecule
- Two ATP molecules are initially used.
- Glucose is split and oxidised into two three-carbon pyruvate molecules.
- Four ATP are formed; therefore, the net gain is two ATP.
- Two NAD^+ accept hydrogen/electrons and form two NADH.

Products per glucose:

2 pyruvate + 2 NADH + net 2 ATP + water

Link reaction and Y — Krebs cycle

Before entering the cycle, each pyruvate enters the mitochondrion and is converted into acetyl-CoA. Carbon dioxide is removed and NADH is formed.

- **Site of Krebs cycle:** Mitochondrial matrix
- Two acetyl-CoA molecules enter per glucose.
- Acetyl-CoA combines with a four-carbon compound.
- A series of reactions regenerates the four-carbon compound.
- Carbon dioxide is released.
- NADH and FADH_2 carry high-energy electrons to the next stage.

Products from the Krebs cycle per glucose:

4 CO_2 + 6 NADH + 2 FADH_2 + 2 ATP

Including the link reaction:

An additional 2 CO_2 and 2 NADH are produced.

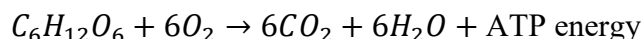
Z — Electron transport chain

- **Site:** Inner mitochondrial membrane/cristae
- NADH and FADH_2 donate electrons to electron carriers.
- As electrons move through the chain, their energy pumps hydrogen ions across the inner membrane.
- Hydrogen ions flow back through ATP synthase.

- ATP synthase forms a large amount of ATP.
- Oxygen is the final electron acceptor and combines with electrons and hydrogen ions to form water.

In the traditional school-level calculation, the complete oxidation of one glucose molecule is described as producing approximately **36–38 ATP**, although the exact yield in living eukaryotic cells may vary.

Overall equation



Question 8 — OR

(a) Leaf fall and darkening of the woody stem

Shedding of yellow leaves

Plants store some wastes in leaves. Before leaf fall, useful substances are withdrawn while waste such as excess salts, calcium oxalate and unwanted metabolic products remain. Chlorophyll breaks down, making the leaves yellow. Formation of an abscission layer causes the leaves to fall, removing the accumulated wastes.

Dark and hard central region of a woody stem

Older xylem vessels stop conducting water and form **heartwood**. Resins, gums, tannins and other waste materials are deposited in these cells. These deposits make the central wood darker, harder and more resistant to decay. Thus, plants isolate wastes in tissues that are no longer metabolically active.

(b) Wilting on a hot sunny day and plant survival

On a hot sunny day:

1. Temperature increases evaporation from mesophyll-cell surfaces.
2. Warm, dry air maintains a steep water-vapour gradient.
3. Stomata may initially remain open for photosynthesis.
4. Transpiration becomes faster than water absorption by roots.
5. Leaf cells lose water and turgor pressure.
6. The leaves and young stems consequently wilt.

Plants survive by:

- Closing stomata through guard-cell action

- Developing a thick waxy cuticle
- Rolling or folding leaves
- Reducing leaf area or producing spines
- Possessing hairs or sunken stomata
- Developing extensive root systems
- Storing water in succulent tissues
- Shedding leaves during severe water shortage

These adaptations reduce water loss and help restore the balance between water absorption and transpiration.